PROMOTING LITERACY · MODULE 2

Every Class Is English Class

Why literacy belongs in 7th grade math, and a plan for PRA

Spencer Bowen · 7th grade math and science
Palm River Academy, Hội An, Vietnam

WHY IT MATTERS IN EVERY GRADE

Reading isn't finished in elementary school

How well students read strongly predicts how well they do in math and science. A student who can't read the question well can't show what they know.

In 2007, **69% of U.S. 8th graders** read below the proficient level for their grade, and 26% read below basic.

The first four recommendations in the IES adolescent literacy guide can be used by **any classroom teacher, in any subject**. Literacy is everyone's job, including the math teacher's.

Kamil et al., 2008, pp. 4, 7 · read the guide

Sounds, then words, then meaning

Phonology

Hearing and saying the sounds of English, and linking them to letters so words can be decoded.

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Language development

Knowing the words and how sentences fit together, including academic English.

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Reading achievement

Understanding grade-level text, like a word problem, well enough to act on it.

Some English learners struggle with the **foundations**: phonological awareness and decoding. They need help with those first.

Another group, often a bigger one, **decodes fine but struggles to understand grade-level text**. They need vocabulary, comprehension and listening work.

My students mostly sit in the second group. Their everyday English is close to native. The little academic details are what catch them out, so that's where my teaching goes.

Baker et al., 2014, p. 61

EXAMPLE 1 · SAYING THE MATH

They knew the signs. They hadn't said them in English.



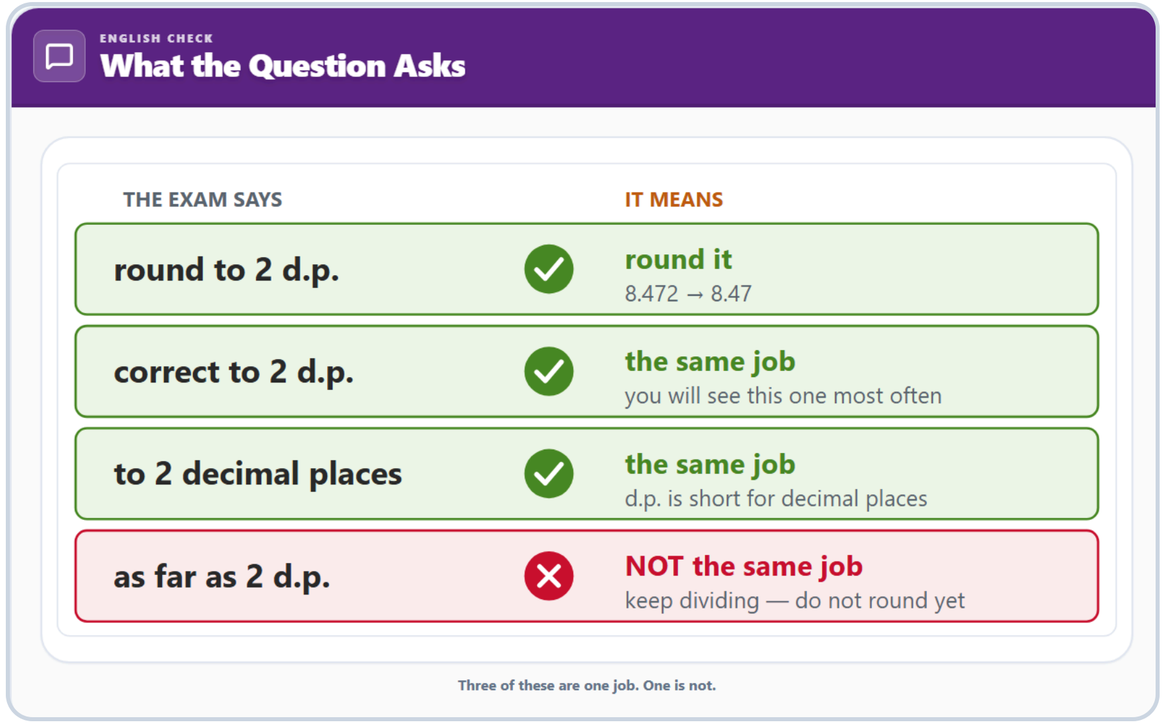
My class has known < and > since first grade in Vietnam. What they hadn't done was **read them aloud in English, left to right**. So my inequalities lesson starts there, before any new math. And “greater” isn't “great.”

Same with powers. A student who has only ever seen 5^2 on paper can't answer when the question is read aloud, so we practice saying **squared** and **cubed**.

My lessons: Inequalities, slide 2 · Square Roots and Cube Roots

EXAMPLE 2 · SMALL WORDS, REAL MARKS

One phrase can change the whole question



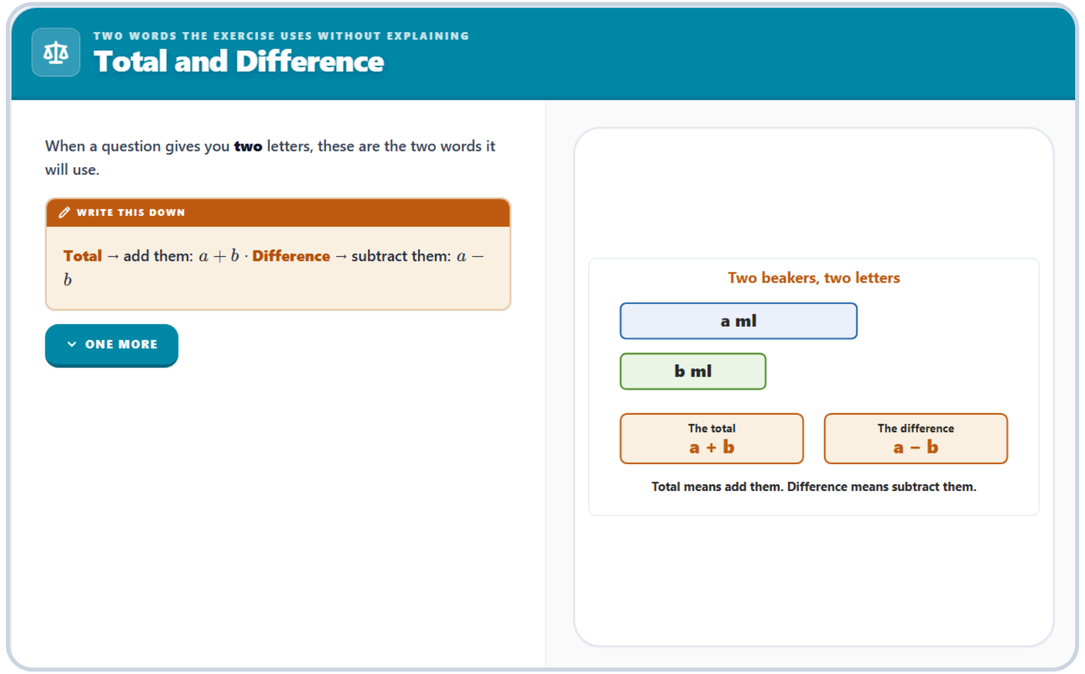
“Correct to 2 decimal places” means round. “As far as 2 decimal places” means keep dividing and don't round yet. It's how the exam asks, and about half the marks in that section depend on noticing the difference.

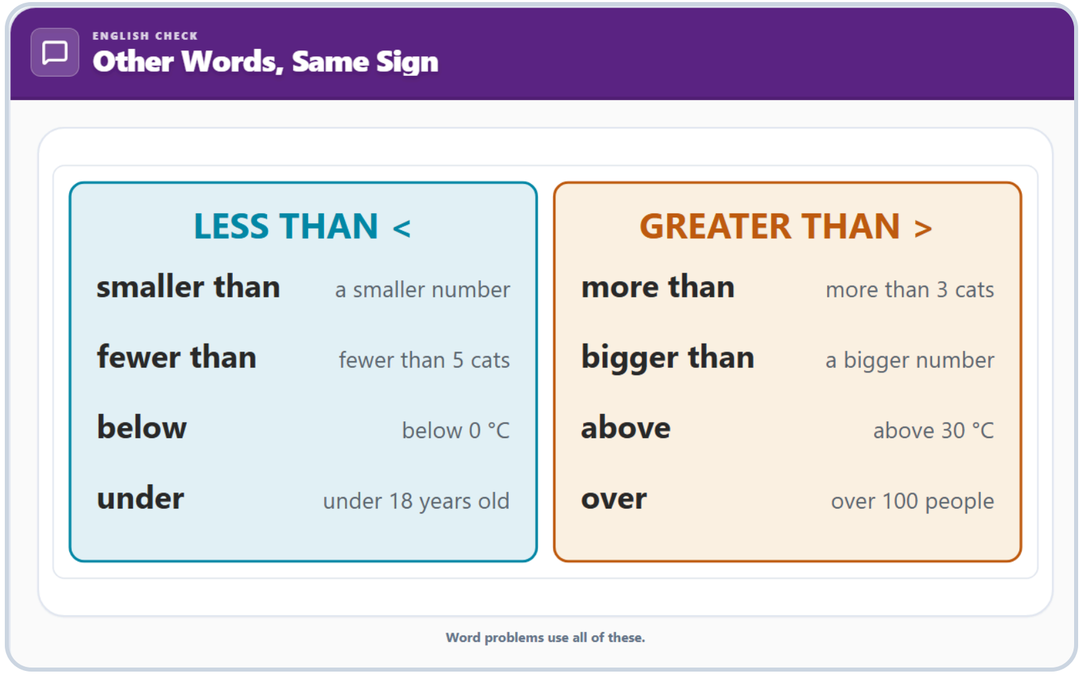
A student who can round perfectly still loses those marks if they misread the instruction.

My lesson: Place Value and Rounding, slide 20

EXAMPLE 3 · WORDS THE BOOK NEVER EXPLAINS

The workbook assumes the English





The exercise uses **total** and **difference** without ever defining them. A student who reads “difference” as “the answer” loses the mark.

Word problems don't always say “less than.” They say **fewer, below, under**. “Under 18” means age < 18.

My lessons: Constructing Expressions, slide 11 · Inequalities, slide 4

SIGNS OF A LITERACY NEED IN 7TH GRADE MATH

What I watch for

- Gets the calculation right and the word problem wrong.
- Answers a different question from the one asked, like rounding when the question said “as far as.”
- Can do 5^2 on paper but not when the question is read aloud.
- Picks out the numbers and skips the words in between.
- Mixes up key words that sound or look alike.
- Can explain the idea in their first language, but stalls when asked to explain it in English.

None of these mean a student is weak at math. Each one points at a reading or language skill I can teach.

PLAN OF ACTION FOR PALM RIVER ACADEMY

Literacy First, in every subject

1 Name the key words

Every subject lists the key words for each unit and teaches them on purpose, over several days, not just once.

Kamil et al., 2008, Rec. 1 · Baker et al., 2014, Rec. 1

2 Say it, then write it

Students read questions aloud and answer in full English sentences in every class, math included.

Baker et al., 2014, Rec. 2

3 Check reading early

A short reading check at the start of the year shows who needs decoding help and who needs vocabulary and comprehension.

Baker et al., 2014, p. 61

4 Small groups for those still stuck

Students who keep struggling get regular small-group help, ideally from someone trained in reading.

Kamil et al., 2008, Rec. 5 · Baker et al., 2014, Rec. 4

5 Teachers learn from each other

Short training on the reading demands of each subject, starting with the words each department finds hardest.

Kamil et al., 2008, p. 7

6 Bring families in

The week's key words go home in English and Vietnamese with my Friday update, so parents can ask about them at home.

from my weekly lesson plan

This term: steps 1, 2 and 6 in my own math and science classes.

Next term: take steps 3 to 5 to PRA's leadership as a plan for the whole academy.

RESOURCES

Where to read more

Improving Adolescent Literacy

Five recommendations for grades 4–12, four of them for any classroom teacher.

ies.ed.gov · Kamil et al., 2008

Teaching Academic Content and Literacy to English Learners

Vocabulary, oral and written English in content classes, and small-group help.

ies.ed.gov · Baker et al., 2014

My Inequalities lesson

Starts with reading < and > aloud in English.

bowenpra.github.io/classroom

My Place Value and Rounding lesson

Slide 20: what the question asks.

bowenpra.github.io/classroom

REFERENCES

Sources

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AI use: I used Claude (an AI assistant) to find sources and build these slides from my own lessons; I reviewed and edited them.